

6.2 Operátor f na prostoru $V = \text{span} \{(2, 1, 0, 1)^T, (1, 0, 1, 2)^T\}$

$$f(u_1) = (4, 1, 2, 5)^T \quad f(u_2) = (2, 2, -2, -2)^T$$

Vyjádříme $f(u_1)$ a $f(u_2)$ jako lin. komb. u_1, u_2

$$f(u_1) \quad a u_1 + b u_2 = (4, 1, 2, 5)^T$$

$$a=1 \quad b=2 \quad [f(u_1)]_B = (1, 2)^T$$

$$f(u_2) \quad c u_1 + d u_2 = (2, 2, -2, -2)^T$$

$$c=2 \quad d=-2 \quad [f(u_2)]_B = (2, -2)^T$$

$$[f]_B = \begin{pmatrix} 1 & 2 \\ 2 & -2 \end{pmatrix}$$

$$\det([f]_B - \lambda I) = (1-\lambda)(-2-\lambda) - 4 = \lambda^2 + \lambda - 6$$

$$\text{Vlastní čísla jsou } \lambda_1 = 2 \quad \lambda_2 = -3 \quad (\lambda+3)(\lambda-2) = 0$$

$$\text{Vlastní vektory: } \parallel \begin{pmatrix} 1 & 2 \\ 2 & -2 \end{pmatrix} - \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$$

$$\text{pro } \lambda_1 = 2: \begin{pmatrix} -1 & 2 \\ 2 & -4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow v_1 = (2, 1)^T_B$$

$$\text{pro } \lambda_2 = -3: \begin{pmatrix} 4 & 2 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow v_2 = (1, -2)^T_B$$

$$w = (5, 1, 3, 7)^T = \alpha u_1 + \beta u_2$$

$$\alpha = 1 \quad \beta = 3 \Rightarrow [w]_B = (1, 3)^T$$

$$\begin{pmatrix} 1 \\ 3 \end{pmatrix} = c_1 \begin{pmatrix} 2 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

$$2c_1 + c_2 = 1$$

$$c_1 - 2c_2 = 3$$

$$c_1 = 3 + 2c_2$$

$$6 + 5c_2 = 1$$

$$c_2 = -1$$

$$c_1 = 1$$

$$[w]_B = 1v_1 - v_2$$

$$[f^n(w)]_B = 1 \cdot \lambda_1^n v_1 - 1 \cdot \lambda_2^n v_2 = 2^n \begin{pmatrix} 2 \\ 1 \end{pmatrix} - (-3)^n \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

$$= \begin{pmatrix} 2 \cdot 2^n - (-3)^n \\ 2^n + 2 \cdot (-3)^n \end{pmatrix}$$

$$f^n(w) = (2 \cdot 2^n - (-3)^n) u_1 + (2^n + 2(-3)^n) u_2$$

$$f^n(w) = (2^{n+1} - (-3)^n) \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} + (2^n + 2 \cdot (-3)^n) \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix}$$

$$f^n(w) = \begin{pmatrix} 2 \cdot 2^{n+1} - 2 \cdot (-3)^n + 2^n + 2 \cdot (-3)^n \\ 2^{n+1} - (-3)^n \\ 2^n + 2 \cdot (-3)^n \\ 2^{n+1} - (-3)^n + 2 \cdot 2^n + 4 \cdot (-3)^n \end{pmatrix} = \begin{pmatrix} 2 \cdot 2^n + 2^n \\ 2 \cdot 2^n - (-3)^n \\ 2^n + 2 \cdot (-3)^n \\ 4 \cdot 2^n + 3 \cdot (-3)^n \end{pmatrix}$$